



CANDIDATE
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REGISTRATION
NUMBER

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H1 Biology

8876/02

Paper 2

15 September 2021

Structured Questions and Free Response Question

2 hours

Additional Materials: 12-page Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Candidates are to answer:

Section A on the Question Paper.

Section B on separate 12-page Answer Booklet.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7 OR 8	
Total	60

This document consists of **16** printed pages and **2** blank pages.

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Answer **all** questions.

- 1** Hormones are responsible for regulating the permeability of kidney cells to water and dissolved ions.

Kidney cells in the collecting duct are impermeable to water. To allow water molecules to move into cells, the hormone vasopressin upregulates the expression of aquaporin channel proteins.

Fig. 1.1 shows the cell surface membrane of a kidney cell, with five proteins labelled **A** to **E**.

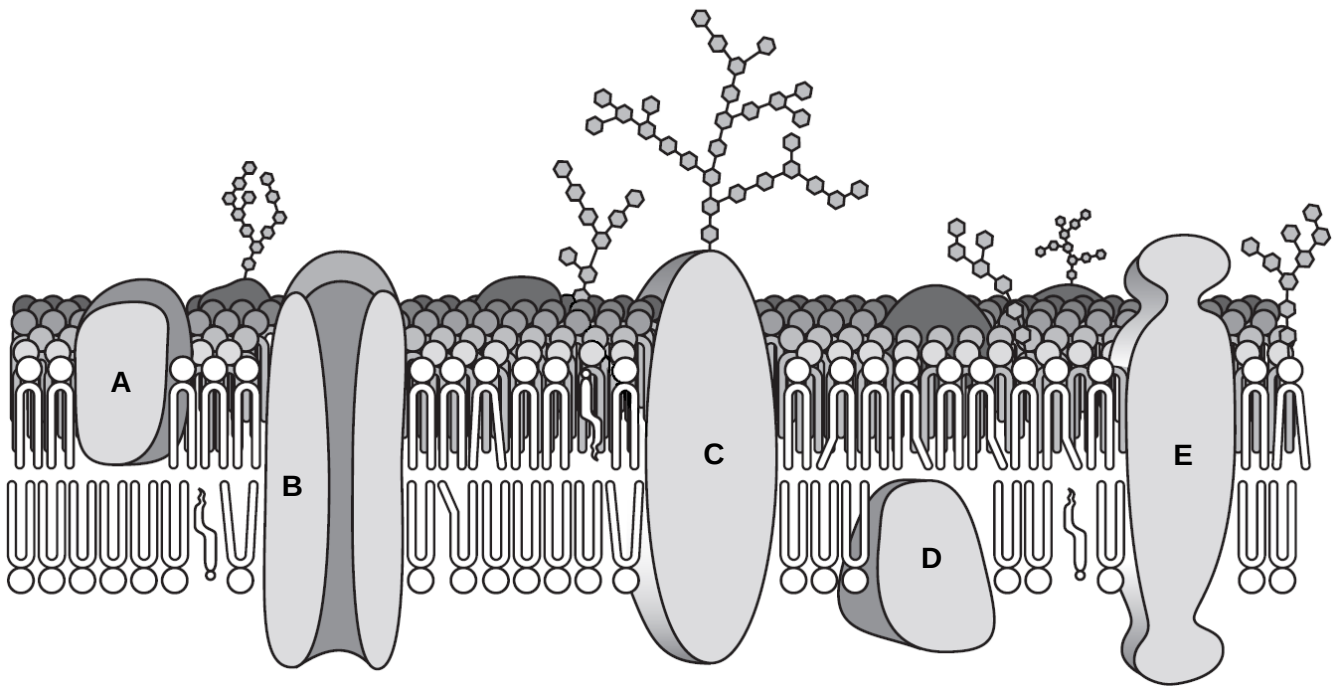


Fig. 1.1

(a) From Fig. 1.1,

- (i)** identify which protein represents the structure of an aquaporin. [1]

B

- (ii)** draw an arrow on Fig. 1.1 to show the direction of movement of water molecules. [1]

Arrow from top to bottom

- (iii)** explain how the arrangement of molecules supports the fluid mosaic model. [2]

1. 'Fluid' = phospholipid and protein molecules are free to move laterally within a layer;
2. 'Mosaic' = random arrangement of the proteins embedded amongst the phospholipid molecules resemble a mosaic pattern;

Vasopressin receptors are G-protein coupled receptors with a transmembrane domain.

(b) Explain how vasopressin receptors are held in the membrane. [2]

1. Phospholipids have a charged and hydrophilic phosphate head which will form hydrogen bonds and ionic bonds with polar and (oppositely) charged amino acids in the protein;
2. The hydrophobic core/fatty acid tails forms hydrophobic interactions with non-polar amino acids in these proteins;

The lipid-soluble hormone aldosterone regulates the level of sodium and potassium ions. It achieves this by upregulating the expression of Na⁺-K⁺ pumps. Fig. 1.2 shows how these protein pumps facilitate the transport of sodium ions into the cell and potassium ions out of the cell.

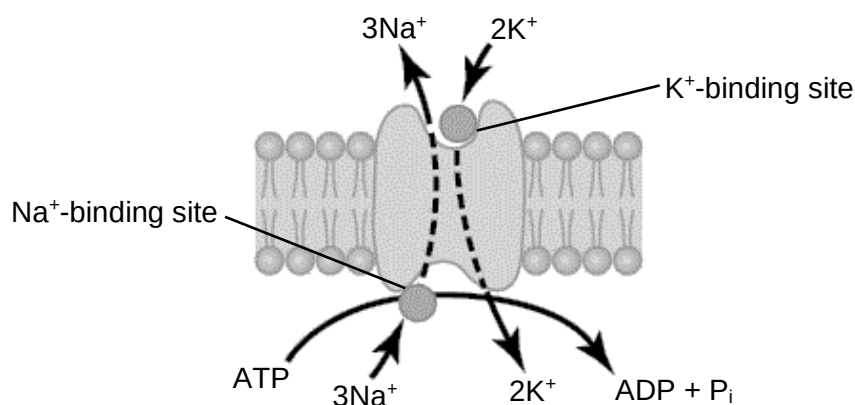


Fig. 1.2

(c) Explain why Na⁺-K⁺ pumps allow Na⁺ and K⁺ to move across the cell surface membrane. [2]

1. Na⁺ and K⁺ ions are charged and hydrophilic, and hence, will be repelled by the hydrophobic core of the cell surface membrane;
2. Instead, they bind to binding sites on the Na⁺-K⁺ pump that will allow the ions to move from one side to the other;

(d) Describe **one** difference in the transport of sodium ions and water molecules into kidney cells in response to aldosterone and vasopressin respectively. [1]

1. Na⁺ ions are pumped into the cell via **active transport**, while water molecules move into the cell via **facilitated diffusion**;
2. Transport of Na⁺ ions into the cell occurs via **Na⁺-K⁺ pumps/protein pumps**, while water molecules diffuse through **aquaporin/channel protein**;
3. Active transport of Na⁺ ions into the cell **utilises energy** released from **ATP hydrolysis** (I: active), while facilitated diffusion of water required **no energy/ATP** (I: passive);
4. Na⁺ ions must bind to the Na⁺-K⁺ pump/protein pump in order to be transported in, while water molecules do not bind to the aquaporin/channel protein;
5. Protein pumps must undergo a change in conformation to release bound Na⁺ ions, while channel proteins do not undergo any conformational change;

[Total: 9]

- 2 Tuberculosis (TB) is an infectious disease caused by the pathogenic bacterium *Mycobacterium tuberculosis*. In this bacterial species, an enzyme known as fructose 1,6-bisphosphate aldolase (FBA) is critical for glucose metabolism.

Fig. 2.1 shows the relative activity of *M. tuberculosis* FBA measured at different pH levels.

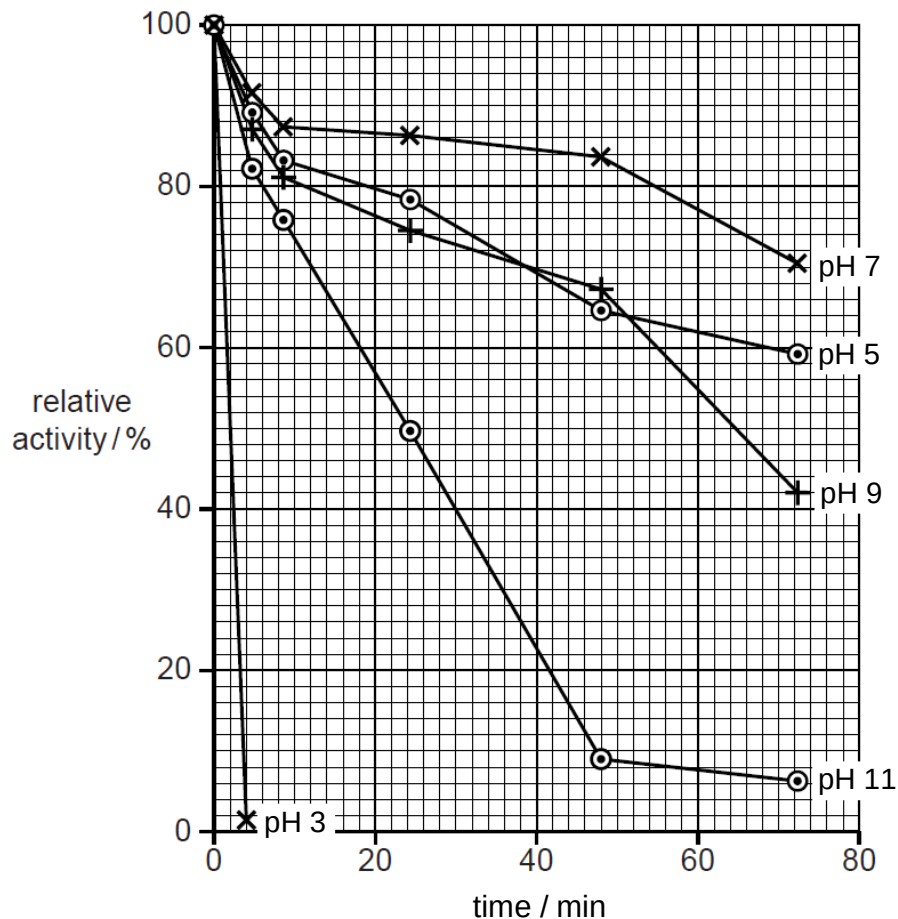


Fig. 2.1

(a) With reference to Fig. 2.1, distinguish between the effect of neutral and alkaline pH on the activity of FBA. [1]

1. From 0 to 72 min, there was a much greater decline (A: more drastic decline) in relative activity of FBA of 93.5% at pH 11 (A: 58.0% at pH 9) as compared to only 29.5% at pH 7;
OR
2. By 72 min, there was greater FBA relative activity remaining at 70.5% at pH 7 as compared to only 6.0-6.5% at pH 11 (A: 42.0% at pH 9);

Fig. 2.2 shows the structure of FBA, which is made of four identical polypeptide subunits.

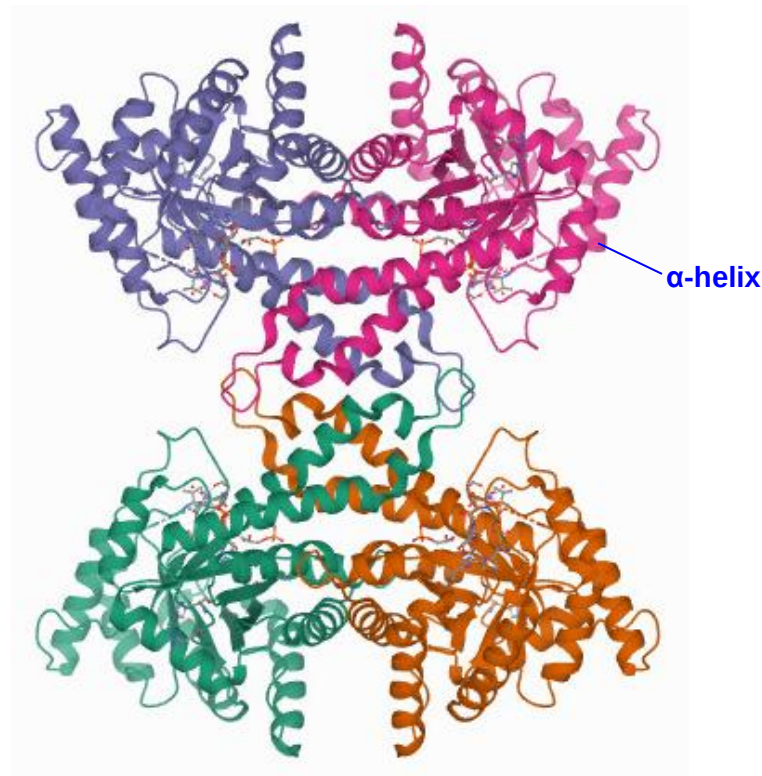


Fig. 2.2

(b) (i) On Fig. 2.2, label and name a secondary structure. [1]

(ii) Explain how changing the pH level affects the structure and activity of FBA. [4]

1. At optimal pH FBA activity is at maximum, and any deviation away from optimal pH would result in sharp decrease in FBA activity;
2. Because at optimal pH, the 3D conformation of FBA's active site is complementary to substrate / fructose 1,6-bisphosphate;
3. However, changes in pH / H^+ concentration can disrupt / break hydrogen and ionic bonds between R groups, hence causing a change in 3D conformation of the active site;

And hence, FBA is denatured as active site is no longer complementary in conformation and charge to substrate molecules; (R: active site is/bonds are denatured)

[Total: 6]

3 Fig. 3.1 shows the normal differentiation pathway of a spermatogonial stem cell into spermatids.

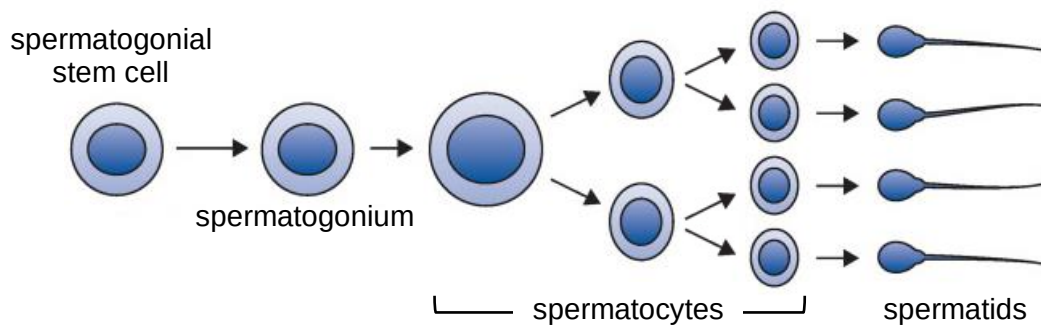


Fig. 3.1

Fig. 3.2 shows three groups of spermatocytes, **A**, **B** and **C**, in different stages of cell division. The peak relative mass of DNA is indicated for each group of cells.

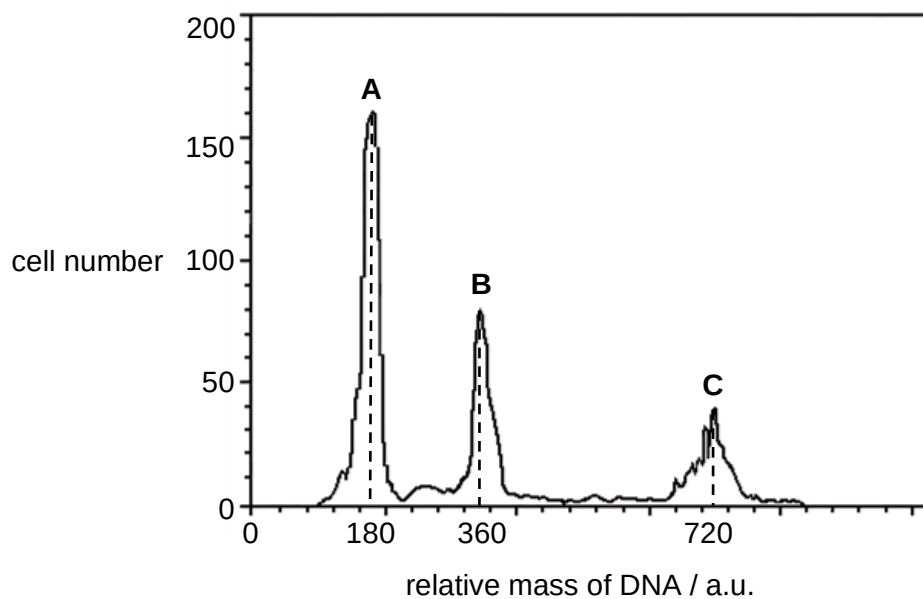


Fig. 3.2

(a) With reference to Fig. 3.2,

(i) identify the group of spermatocytes that are formed at the end of

[1]

meiosis I	meiosis II
B	A

(all or none)

(ii) explain your answer in (a)(i). [3]

1. Spermatocyte undergoes meiosis I and meiosis II, where relative mass of DNA is halved from 720 to 360 a.u. and halved again from 360 to 180 a.u. respectively;
2. Because pairs of homologous chromosomes are first separated into two daughter cells at the end of meiosis I;
3. And subsequently, sister chromatids (which are now daughter chromosomes) are separated into two gametes/spermatids at the end of meiosis II;

Germ cell tumours result from uncontrolled division of germ cells, which give rise to gametes. In males between the ages of 15 to 35 years, the risk of testicular cancer increases with age.

Fig. 3.3 shows how a tumour may result from uncontrolled cell division of a spermatogonium (male germ cell), instead of normal differentiation into spermatids.

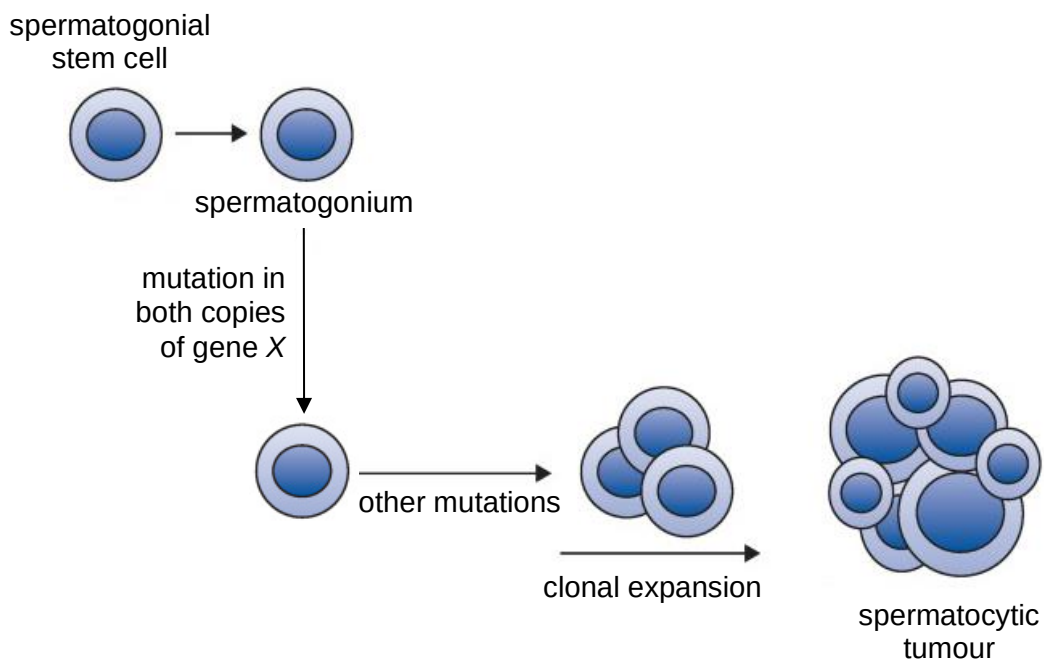


Fig. 3.3

(b) (i) Suggest the name of gene X. [1]

p53

(ii) Explain how gene X may result in uncontrolled cell division of a spermatogonium. [3]

1. A loss-of-function mutation of both alleles of p53 / tumour suppressor gene,
2. Which results in failure to arrest cell cycle / activate DNA repair mechanisms;
3. Thus leading to accumulation of (other) mutations in a single cell, which leads to dysregulation of cell cycle checkpoints;

A: gain-of-function mutation in proto-oncogenes and other tumour suppressor genes leads to dysregulation of cell cycle checkpoints;

- (c) Cancer treatments, including chemotherapy and radiotherapy, can often result in loss of fertility. In male cancer patients, sperm cryopreservation (freezing of sperm cells) is one method of preserving fertility.

Using your knowledge of stem cells, suggest why fully differentiated sperm cells are preserved instead of spermatogonial stem cells. [1]

Any one:

1. Spermatogonial stem cells are much fewer in number than sperm cells, therefore harder to extract;
2. Spermatogonial stem cells must be induced using appropriate molecular signals to undergo differentiation into specialised sperm cells, which requires an additional step;
3. Any existing mutations within a spermatogonial stem cell will be passed on to all sperm cells that result from meiotic division;
4. AVP;

[Total: 9]

- 4 In the bacterium *Thermus thermophilus*, a sequence of DNA contains two overlapping genes, *rnpA* and *rpmH*, which code for RNase P and a ribosomal protein respectively.

Fig. 4.1 shows the first 30 bases in the DNA sequence. Following process X, a different strand of nucleic acid is formed, and it contains the translation start sites for RNase P and the ribosomal protein.

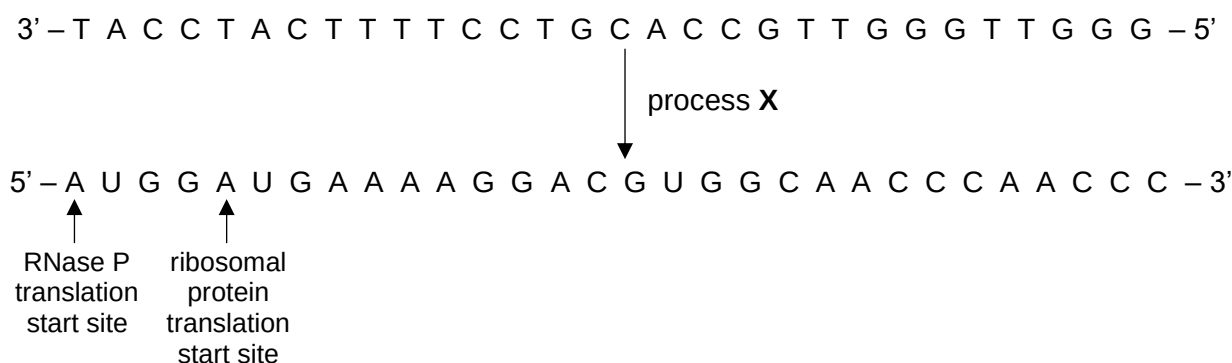


Fig. 4.1

- (a) (i) Briefly describe process X. [3]
1. RNA polymerase binds to the promoter to initiate transcription;
 2. It unzips double stranded DNA, and uses one DNA strand as a template to synthesise an mRNA strand;
 3. Via complementary base pairing between adenine & uracil, cytosine & guanine, thymine & adenine (must have all three);
 4. It catalyses the formation of phosphodiester bonds between ribonucleotides to form the mRNA strand;
 5. mRNA strand is synthesised in the 5' to 3' direction (A: template DNA strand is read in the 3' to 5' direction);
- (ii) Suggest why the product of process X can code for two different polypeptide products. [2]
1. After transcription of this DNA sequence, there are two start codons AUG in the mRNA strand;
 2. Ribosome begins translation four nucleotides apart / in different reading frames / reads a different sequence of mRNA codons (OWTTE), hence producing two different amino acid sequences can be produced from the same mRNA strand;

In eukaryotic cells, the *FECH* gene codes for ferrochetalase, which is the final enzyme in the metabolic pathway of haem synthesis.

A study done in zebrafish found that a point mutation in the *FECH* gene resulted in a disease phenotype known as dracula. The disease is characterised by lysis of red blood cells upon exposure to light.

Fig. 4.2 shows a segment of the *FECH* template DNA sequence in normal and dracula zebrafish, and Table 4.1 shows the genetic code.

normal

27	28	29	30	31	32	33
CAC	GTG	GAC	AGT	GGA	CTC	ATC

dracula

27	28	29	30	31	32	33
CAC	GTG	GAC	ATT	GGA	CTC	ATC

Fig. 4.2

Table 4.1

	U		C		A		G	
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys
	UUC		UCC		UAC		UGC	
	UUA	Leu	UCA		UAA	STOP	UGA	STOP
	UUG		UCG		UAG	STOP	UGG	Trp
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg
	CUC		CCC		CAC		CGC	
	CUA		CCA		CAA	Gln	CGA	
	CUG		CCG		CAG		CGG	
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser
	AUC		ACC		AAC		AGC	
	AUA	Met	ACA		AAA	Lys	AGA	Arg
	AUG		ACG		AAG		AGG	
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly
	GUC		GCC		GAC		GGC	
	GUA		GCA		GAA	Glu	GGA	
	GUG		GCG		GAG		GGG	

(b) Using Fig. 4.2 and Table 4.1,

(i) State the amino acid sequence for this segment of the normal polypeptide. [1]

Val – His – Leu – Ser – Pro – Glu – Met

(ii) Explain how the point mutation results in non-functional ferrochetalase. [2]

1. The point substitution mutation, where G is replaced with T, introduces a premature stop codon UAA (instead of UCA - Serine), thus leading to a nonsense mutation;
2. Translation is prematurely terminated, resulting in a truncated polypeptide, which then folds into a protein of different 3D conformation that is non-functional;

[Total: 8]

- 5 Marfan syndrome is a genetic disorder that affects the connective tissue. Fig. 5.1 shows the inheritance pattern of Marfan syndrome in a family.

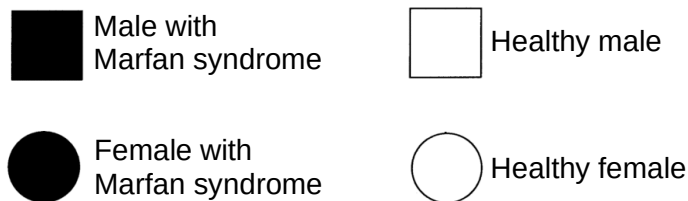
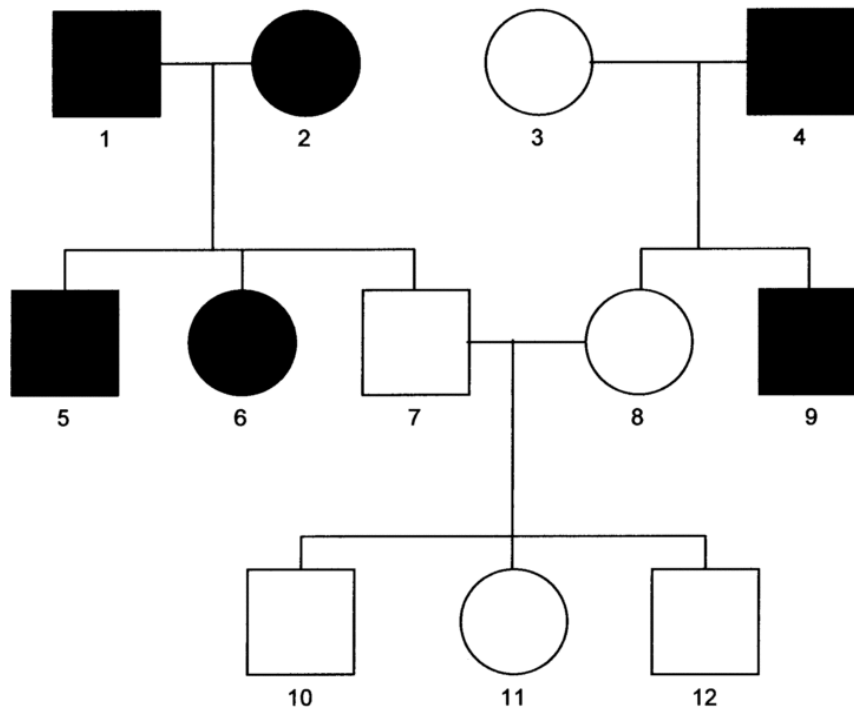


Fig. 5.1

- (a) (i) State if Marfan syndrome is caused by a recessive or dominant allele. [1]
Dominant allele;

- (ii) Explain your answer to (a)(i). [2]

Marfan syndrome is caused by a dominant allele because

1. individuals 1 and 2 suffer from Marfan syndrome and yet have a healthy son;
2. This means that the normal allele is masked by the allele that causes the Marfan syndrome in individuals 1 and 2;

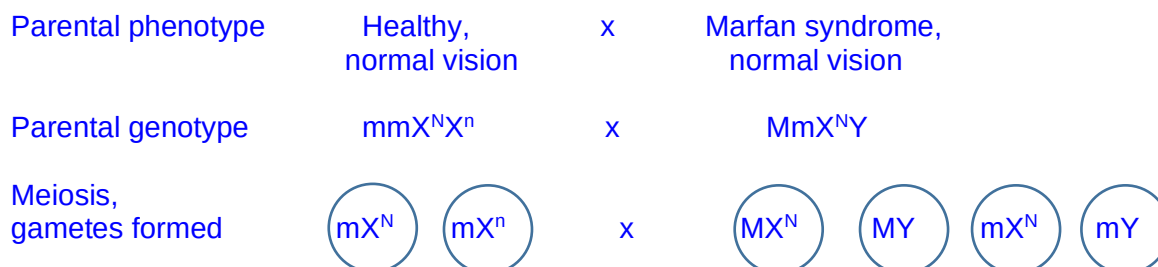
- (b) Other than the inheritance of Marfan syndrome, the inheritance of colour blindness was also tracked for the same family. For instance, individuals 3 and 4 have normal vision but their son (individual 9) has colour blindness. Colour blindness is a sex-linked trait.

Based on the mode of inheritance of Marfan syndrome and colour blindness, draw a genetic diagram to show the expected offspring phenotypic ratio that could result for the children born to individuals 3 and 4.

Let X^N represent the X chromosome with the normal vision allele and X^n represent the X chromosome with colour blindness allele.

Let Y represent the Y chromosome.

Let M represent the Marfan syndrome allele and m represent the normal allele.



Fertilisation

Gametes	MX^N	MY	mX^N	mY
mX^N	MmX^NX^N Marfan syndrome, normal vision	MmX^NY Marfan syndrome, normal vision	mmX^NX^N Healthy, normal vision	mmX^NY Healthy, normal vision
mX^n	MmX^NX^n Marfan syndrome, normal vision	MmX^nY Marfan syndrome, colour blind	mmX^NX^n Healthy, normal vision	mmX^nY Healthy, colour blind

Offspring genotypic ratio 3 $M_X^N_$: 1 MmX^nY : 3 $mmX^N_$: 1 mmX^nY

Offspring phenotypic ratio 3 Marfan syndrome, normal vision : 1 Marfan syndrome, colour blind : 3 healthy, normal vision : 1 healthy, colour blind

[4]

[Total: 7]

- 6 The coho salmon, *Oncorhynchus kisutch*, is a species of North American salmon. Adult salmon spend majority of their time in the salt water of the Pacific Ocean until it is time for them to migrate back to the natal stream. The natal stream is where the salmon first hatched from eggs, and they return to the natal stream as adults to spawn (i.e. release sperm or eggs for mating).

At the natal stream, adult male salmon engage in physical fights to gain access to female, with larger sized males often emerging as winners and earning the opportunity to mate with females. Some adult males, however, are much smaller in size. These males are known as 'jacks'. Jacks hide around smaller boulders in the streambed to avoid fighting with bigger males. As part of his mating strategy, the jack "sneaks" in close to a female to release his sperms just as the female is depositing her eggs. This increases his chance to fertilize the females' eggs without needing to fight with other bigger males. In this way, both jacks and normal breeding adult males are able to spawn successfully.

Fig. 6.1 shows the variation in body length of reproductively mature males in an original population of coho salmon, before the evolution of the jack phenotype in the population.

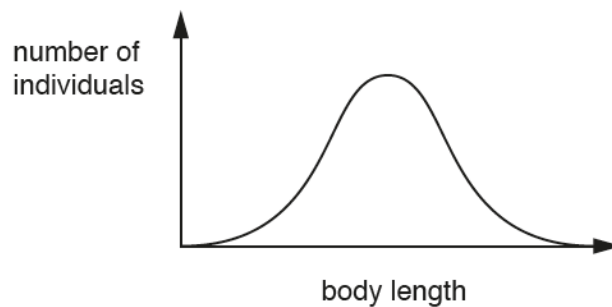


Fig. 6.1

Selection has since acted over time on this original population to change the distribution of body length in reproductively mature male salmon.

- (a) (i) Sketch on Fig. 6.1 to show the new distribution of body length in the present-day population of reproductively mature male salmon. [1]

Two peaks, either side of the normal peak ;

- (ii) Name the type of selection that is occurring. [1]

Disruptive selection;

In Chilean salmon, the bacteria *Piscirickettsia salmonis* causes the disease piscirickettsiosis, which can have a major impact on farmed salmon populations. To combat this, the antibiotic oxytetracycline is most frequently used to target *P. salmonis* outbreaks.

Table 6.1 shows the percentage of oxytetracycline resistant *P. salmonis* strains in a population of farmed Chilean salmon. The percentage of antibiotic resistant bacterial strains was assessed at 1, 2, 12 and 18 months after the start of oxytetracycline therapy.

Table 6.1

time after start of therapy / months	antibiotic resistant bacterial strains / %
1	2
2	10
12	20
18	50

(b) With reference to Table 6.1, explain how the use of oxytetracycline therapy can lead to the evolution of antibiotic resistant bacterial strains. [4]

1. [Description of data]: With increasing time in therapy, the percentage of AZT resistant strains in the patients increases from 2% in month 1 to 50% in month 18 (need to quote data).
2. Mutations in HIV led to AZT resistance in the population hence leading to variation;
3. AZT acts as selection pressure and AZT-resistant strains are selected for;
4. HIV strains which are AZT-resistant survive and reproduce at a higher rate and pass on the favourable alleles to newly formed viral progeny;
5. Over time, the frequencies of alleles which confer resistance to AZT increase, leading to increase in the percentage of resistant strains as seen in Table 9.1.

[Total: 6]

Section B

Answer **one** question in this section.

Write your answers on the 12-page answer booklet provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts (a), (b) etc., as indicated in the question.

Either

- 7 (a) Explain the importance of non-covalent bonds in cellular processes. [9]

(G) Role of non-covalent bonds in DNA replication / Gene expression (max 4m)

1. The two strands in a DNA molecule are held together by hydrogen bonds;
2. This allows for helicase to unzip the DNA molecule so that each DNA strand can be used as a template for DNA replication/synthesis of new DNA strand;
- OR
This allows for RNA polymerase to unzip the DNA molecule so that template DNA strand can be used for synthesis of RNA;
3. Hydrogen bonds are formed between complementary base pairs of codon on mRNA and anti-codon of tRNA;
4. This allows for the specific sequence of amino acids to be joined together in a polypeptide chain during translation;
5. AVP;

(E) Role of non-covalent bonds in enzymatic reactions (max 4m)

1. Substrate binds to active site of enzyme via non-covalent bonds such as hydrophobic interactions, ionic bonds and hydrogen bonds;
2. This allows for enzyme-substrate complex to be formed which lowers activation energy of the reaction;
3. It also allows for products formed to leave the active site easily so that enzyme can be re-used;
4. AVP;

(T) Role of non-covalent bonds in transport processes (max 4m)

1. Weak hydrophobic interactions between phospholipids of cell surface membrane;
2. This results in fluidity of the cell surface membrane and transient pores are formed which allows for the diffusion of substances across the cell surface membrane;
3. Substance to be transported binds to carrier protein / pump on cell surface membrane via non-covalent bonds such as hydrophobic interactions, ionic bonds and hydrogen bonds;
4. This allows for conformation change to occur in carrier protein / pump, and the substance is then transported across the cell surface membrane via facilitated diffusion / active transport respectively;

5. Substance to be transported binds to receptor on cell surface membrane via non-covalent bonds such as hydrophobic interactions, ionic bonds and hydrogen bonds;
6. This allows for receptor –mediated endocytosis to occur to result in uptake of the substance into the cell;
7. AVP;

QWC: answer covers role of non-covalent bonds in two different cellular processes

- (b) Herbicides work in a number of different ways. The herbicide atrazine works by disabling plastoquinone, which is one of the electron carriers in the electron transport chain between photosystem I and II.

Explain how atrazine would kill a susceptible plant.

[6]

(G) Role of non-covalent bonds in DNA replication / Gene expression (max 4m)

1. In the presence of atrazine, movement of electrons down the electron transport chain is prevented;
2. As such, protons are not pumped from stroma to thylakoid space and proton motive force is not generated;
3. Production of ATP via chemiosmosis during photophosphorylation / light-dependent reaction comes to a halt;
4. NADPH is also not produced;
5. Without the ATP and NADPH from the light dependent stage, Calvin cycle also stops;
6. Glyceraldehyde-3-phosphate is not formed and glucose production in the plant ceases;
7. Plant has no respiratory substrate to carry out respiration;
8. AVP (e.g. no ATP produced from respiration for named vital processes in plant);

OR

- 8 (a) Explain the importance of cyclical processes that occur in a plant.

[9]

(P) Role of cyclic photophosphorylation (max 4m)

1. During cyclic photophosphorylation, light energy is absorbed and electron from P700 is emitted;
2. The electron is passed down the electron transport chain which is composed of a series of electron carriers of decreasing energy levels. Energy is released and used to pump protons from the stroma into the thylakoid space, creating a proton motive force;
3. Protons diffuse through ATP synthase and ATP is synthesised via chemiosmosis;

Role:

4. ATP is used during the reduction of glycerate-3-phosphate to glyceraldehyde-3-phosphate / regeneration of RuBP during Calvin cycle;

(C) Role of Calvin cycle (max 4m)

1. During Calvin cycle, Rubisco carries out CO₂ fixation;
2. Glycerate-3-phosphate is formed and is reduced to glyceraldehyde-3-phosphate (G3P) using ATP and NADPH;

Role

3. G3P can be used to form glucose, which can be used for the synthesis of cell wall;
4. Glucose formed can also be used for respiration by the plant for the synthesis of ATP;
5. NADP⁺ is also regenerated, and this allows non-cyclic photophosphorylation / photosynthesis to continue in the presence of light;

(K) Role of Krebs cycle (max 4m)

1. During the Krebs cycle, acetyl-coA is oxidised and this results in the production of 6 NADH and 2 FADH₂ per glucose;
2. NADH and FADH₂ will transport high energy electrons to the electron transport chain at the cristae for oxidative phosphorylation;

Role:

3. Each NADH will result in the production of 3 ATP;
4. Each FADH₂ will result in the production of 2 ATP;
5. 2 molecules of ATP are produced per glucose molecule via substrate-level phosphorylation;

QWC: answer covers role of two different cyclical processes that occur in plants;

- (b)** Water can fill air spaces in the soil surrounding the roots. This prevents oxygen from reaching root hair cells.

Using your knowledge of aerobic and anaerobic respiration, explain why overwatering can kill plants. [6]

Impact of lack of oxygen on aerobic respiration

1. When oxygen does not reach the root hair cells, oxidative phosphorylation will stop as oxygen, the final electron acceptor, is absent;
2. NAD⁺ and FAD are not regenerated and link reaction and Krebs cycle will come to a halt;

Impact of lack of oxygen on anaerobic respiration

3. Anaerobic respiration, which is comprised of glycolysis and alcohol fermentation will proceed;
4. Anaerobic respiration produces only 2 ATP per molecule of glucose unlike 38 ATP per glucose molecule for aerobic respiration;
5. The amount of ATP produced is too little for plant to survive / cannot make DNA, proteins, chlorophyll / cannot carry out active transport of minerals;
6. During alcohol fermentation, pyruvate is converted to ethanol;
7. Accumulation of ethanol is toxic to plant cells;

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